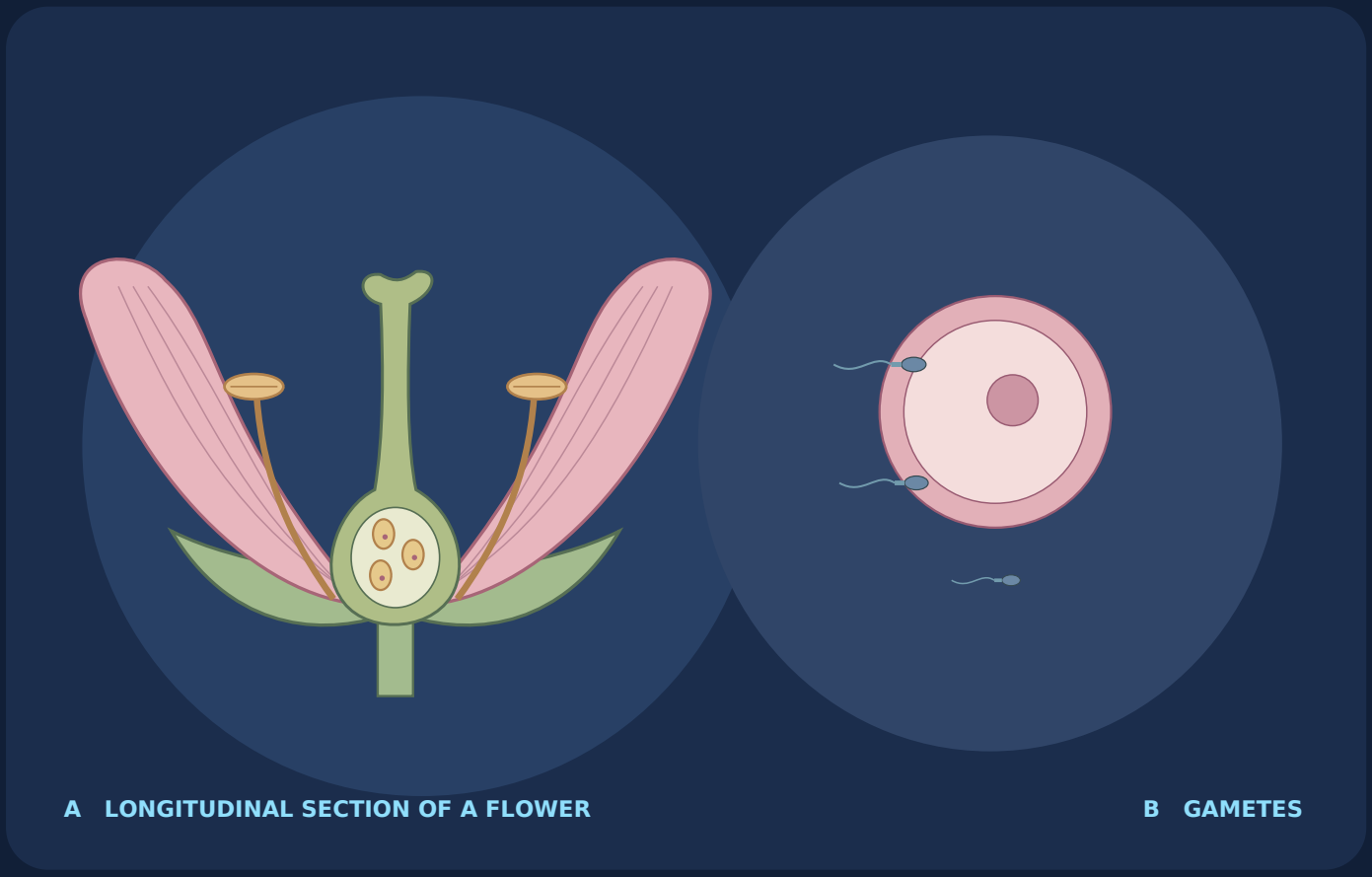
RAMA'S ACADEMY / CLASS 9 BIOLOGY

Reproduction

How Life Continues

EXPANDED STUDENT CHAPTER



A LONGITUDINAL SECTION OF A FLOWER

B GAMETES

READ IN SEQUENCE

Asexual reproduction and propagation

Sexual reproduction in flowering plants and animals

Human reproduction, development and reproductive health

THEORY / LABELLED FIGURES / STUDY QUESTIONS

FOUNDATIONS

Chapter Contents

FOUNDATIONS

Reproduction: How Life Continues 3

ASEXUAL REPRODUCTION

Asexual Reproduction 4

Vegetative Propagation in Agriculture 5

Tissue Culture and Modern Farming 6

Budding in Yeast and Hydra 7

Spore Formation in Fungi 8

SEXUAL REPRODUCTION

Sexual Reproduction 9

FLOWERING PLANTS

Sexual Reproduction in Flowering Plants 10

Pollination 11

Pollination Strategies and Reproductive Success 12

Insect and Bird Pollination 13

Fertilisation and Seed Formation 14

Plant Breeding and Its Applications 15

ANIMALS

Sexual Reproduction in Animals 16

Variations in Reproduction in Animals 17

HUMAN REPRODUCTION

Reproduction in Human Beings 18

Female Reproductive System 19

Gametogenesis 20

Fertilisation and Implantation 21

Menstrual Cycle 22

Curiosity Corner and Menstrual Hygiene 23

Pregnancy and Childbirth 24

Childbirth and Care 25

REPRODUCTIVE HEALTH

Postpartum Health and Sexual Maturity 26

Prevention of STIs and Unintended Pregnancy 27

Family Planning and Community Health 28

REVISION

Mind Map and Key Distinctions 29

PRACTICE

NCERT Folder 30

Exam-Style Questions 31

Self Assessment 32

Answers and Source Notes 33

FOUNDATIONS

Reproduction: How Life Continues

A potato kept in a cupboard can grow shoots. A yeast cell can grow a bud. A flowering plant can produce seeds that travel far from the parent. Each example begins differently, but the result is the same: new individuals of the same species. This process is called **reproduction**.

Why reproduction matters

An individual organism can survive without reproducing. A species can continue across generations only when some of its members produce offspring. In a life cycle, an organism begins life, grows, reaches maturity and eventually dies. Some mature individuals reproduce before they die. Their offspring then begin the next generation. Reproduction therefore links one life cycle to another.

Two main routes

In **asexual reproduction**, a single parent gives rise to offspring without the fusion of reproductive cells. A yeast bud and a potato shoot are examples. In **sexual reproduction**, two reproductive cells called **gametes** fuse. The resulting cell is a **zygote**. In many species the gametes come from different individuals. Some flowering plants can produce both gametes on the same plant.

Inheritance and variation

Offspring inherit genetic information from their parent or parents. In asexual reproduction they generally inherit a near-identical copy. In sexual reproduction they inherit a new combination. The resulting differences are called **variation**. If a disease spreads or the environment changes, a population with varied traits may contain individuals better able to survive and reproduce under the new conditions. Over generations, inherited variation can contribute to adaptation and evolution.

Follow the chapter

We first look at reproduction without gamete fusion. We then study meiosis and sexual reproduction, starting with flowers, moving to animals, and finally tracing the human reproductive cycle and reproductive health.

A LIFE CYCLE LINKS GENERATIONS

New individual → Growth → Maturity → Reproduction

TRY IT

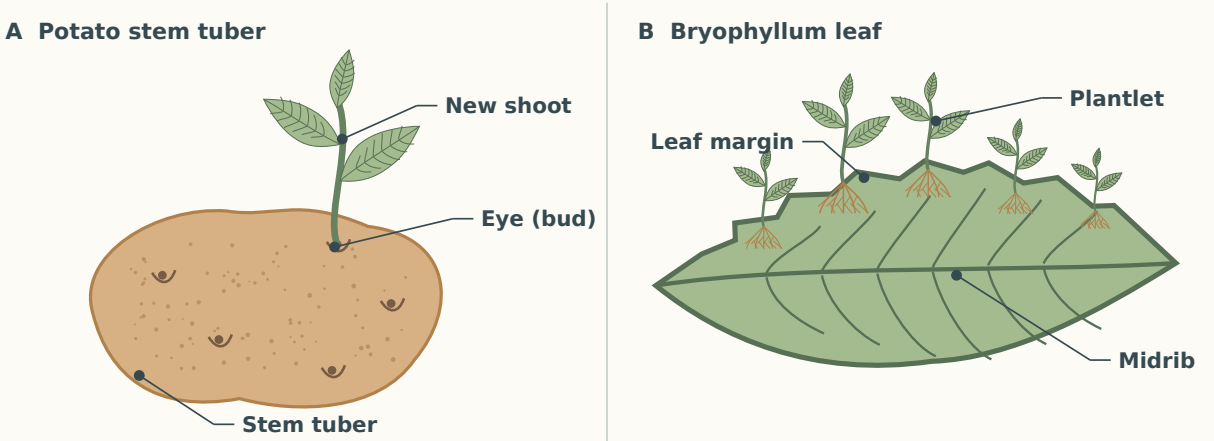
A potato shoot and a seed can both become new plants. Which one starts from a vegetative part of the parent?

ASEXUAL REPRODUCTION

Asexual Reproduction

In asexual reproduction, one parent supplies the genetic information for the next generation. No sperm and egg fuse. In organisms with a nucleus, DNA is copied before cell division. During **mitosis**, the copied chromosomes are distributed to the daughter cells. The offspring are usually genetically very similar to the parent and are called **clones**. A mutation can introduce a difference, and growing conditions can make genetically similar individuals look different.

VEGETATIVE PROPAGATION FROM A STEM AND A LEAF



Both structures can produce a new plant without fertilisation.

Common forms

An Amoeba divides into two individuals by **binary fission**. Some organisms reproduce by **fragmentation**, in which a broken-off piece can grow into a new individual. Yeast and Hydra reproduce by budding, and many fungi produce spores. In plants, a new individual can develop from an ordinary body part.

Vegetative Propagation in Plants

Vegetative propagation starts from a **root, stem or leaf**. A potato is a swollen underground stem, or tuber. The eyes on its surface are buds that can produce shoots. Ginger has a horizontal underground stem called a **rhizome**. A stem piece of money plant or sugarcane can sprout from its nodes. In *Bryophyllum*, small plantlets with shoots and roots form along the leaf margin and can grow independently after detaching.

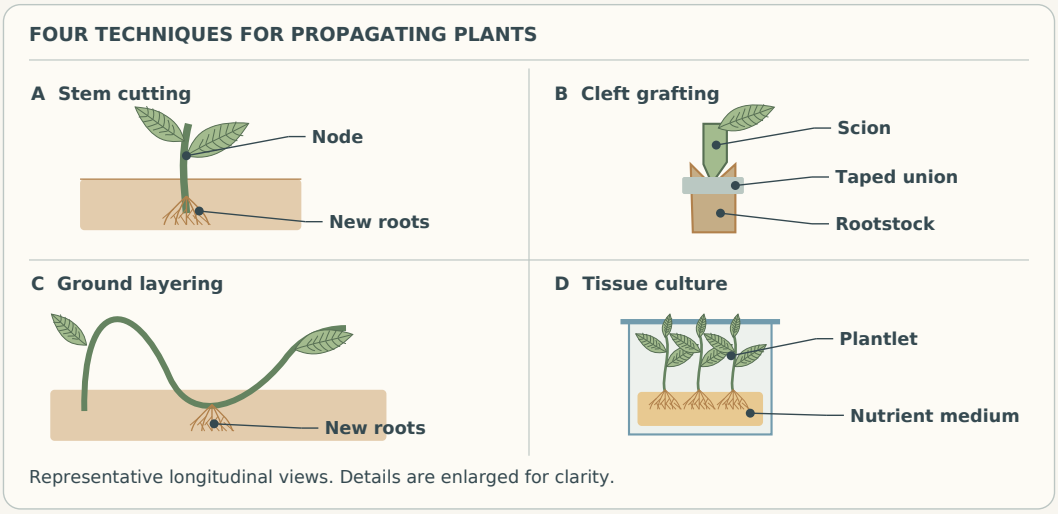
FIGURE CHECK

Compare the eye of a potato with a plantlet on a Bryophyllum leaf. Which plant organ gives rise to each?

ASEXUAL REPRODUCTION

Vegetative Propagation in Agriculture

Farmers and gardeners use vegetative propagation when they want many plants with particular traits, such as a fruit’s taste or a flower’s colour. It can also multiply crops that rarely produce usable seed. Planting many clones has a trade-off. When the clones share a susceptibility to a disease, the disease can affect much of the crop.



Representative longitudinal views. Details are enlarged for clarity.

1. Cutting

A grower removes a healthy stem piece, often with one or more nodes, and places it in moist soil or another rooting medium. Buds develop into shoots while new roots form on the cutting. Rose, money plant and sugarcane are common examples. The detached piece must survive until its roots are able to take up enough water.

2. Grafting

The grower cuts a slit in a rooted plant, the **rootstock**, and fits a wedge-shaped shoot, the **scion**, into it. Their growing tissues must touch so the two parts can heal together. Tape protects the union, and competing shoots below the graft may be removed. The rootstock supplies the root system. The scion produces shoots and fruit with its own traits. Mangoes and apples are often propagated this way.

3. Layering

A flexible branch is bent to the ground and part of it is covered with moist soil. The tip stays above ground. The branch remains attached while roots form, so it continues to receive support from the parent. It is then cut free. Lemon, jasmine and guava can be propagated by layering.

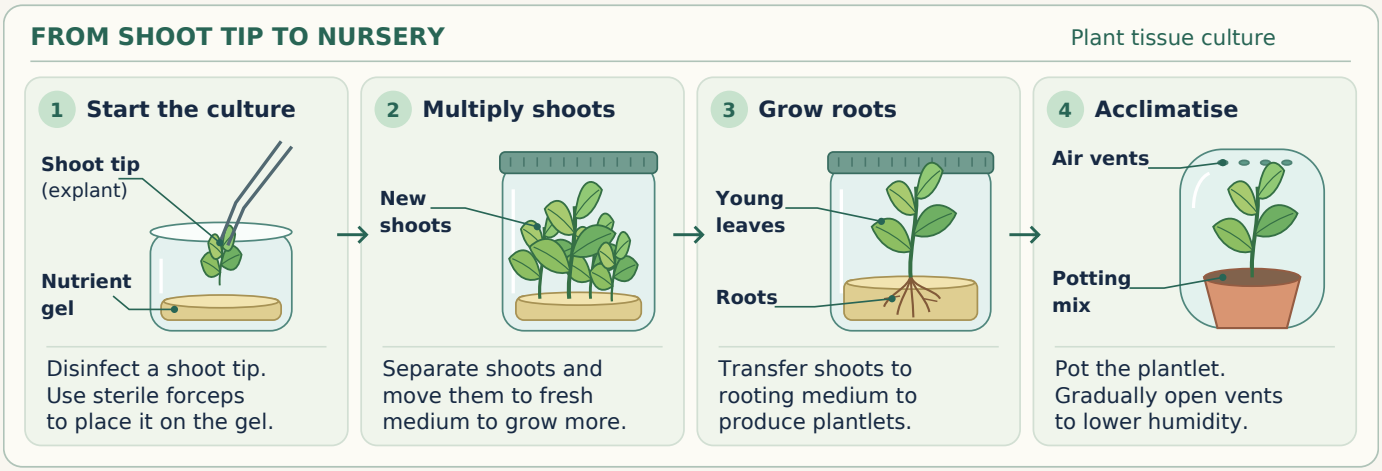
GROWER’S DECISION

To preserve the fruit of one sweet mango tree, take the scion from that tree and graft it onto suitable rootstock.

ASEXUAL REPRODUCTION

Tissue Culture and Modern Farming

Tissue culture, or micropropagation, starts with a small piece of living plant material called an **explant**. A shoot tip is often useful because it includes an **apical meristem**, a region where cells are actively dividing. The explant is transferred to a sterile container with a nutrient medium. Sterility matters because bacteria and fungi can multiply on the same medium and overtake the culture.



From one explant to many plants

The medium supplies sugars and minerals. Plant growth regulators in the medium can help control when shoots and roots form. In a typical procedure, the explant first develops shoots. These shoots can be divided and transferred to fresh medium to increase their number. Shoots then produce roots. Once they have both roots and leaves, the young plants are called **plantlets**.

Moving out of the laboratory

Plantlets raised in a closed container are used to stable humidity and light. Before they go into a nursery or field, they are gradually exposed to ordinary conditions. This stage is called **acclimatisation**. With suitable starting material, screening and careful handling, the method can produce large batches of uniform planting stock and help reduce the spread of some diseases.

Where it helps

Bananas and orchids are examples of plants multiplied by tissue culture. Banana growers can obtain many planting units from selected material, while orchid nurseries can multiply valuable varieties from a small sample. Agricultural researchers, including teams in India's ICAR system, study propagation methods to improve crop production.

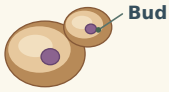
COMPARE METHODS

Layering roots a branch while it is still attached. Tissue culture multiplies small pieces of plant tissue under controlled laboratory conditions.

ASEXUAL REPRODUCTION

Budding in Yeast and Hydra

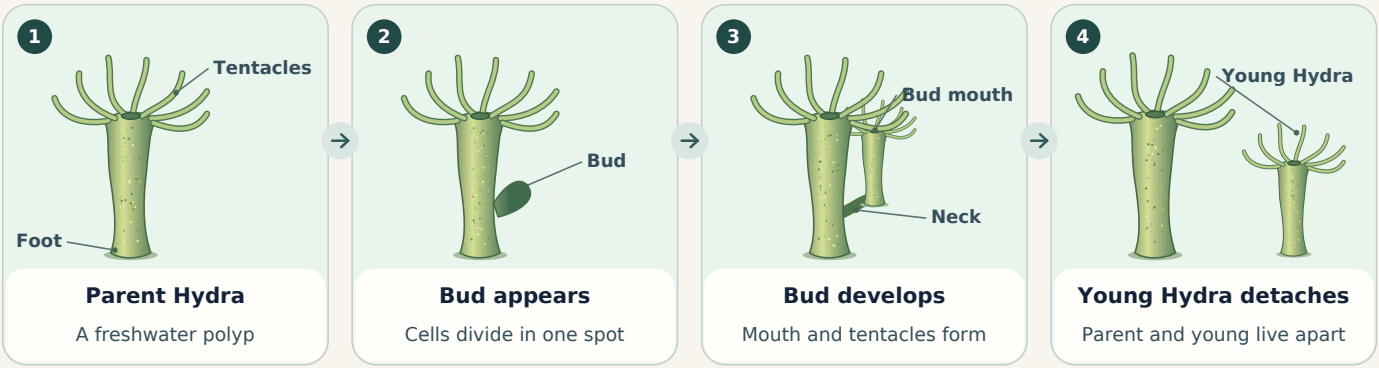
In **budding**, a new individual grows from an outgrowth on one parent. No gametes fuse. In yeast the bud becomes a daughter cell. In Hydra it develops into a complete animal.



Budding in yeast. The nucleus divides, and one daughter nucleus enters the enlarging bud. A partition forms across the neck. The daughter cell can then separate, although attached buds sometimes form a short chain.

Hydra

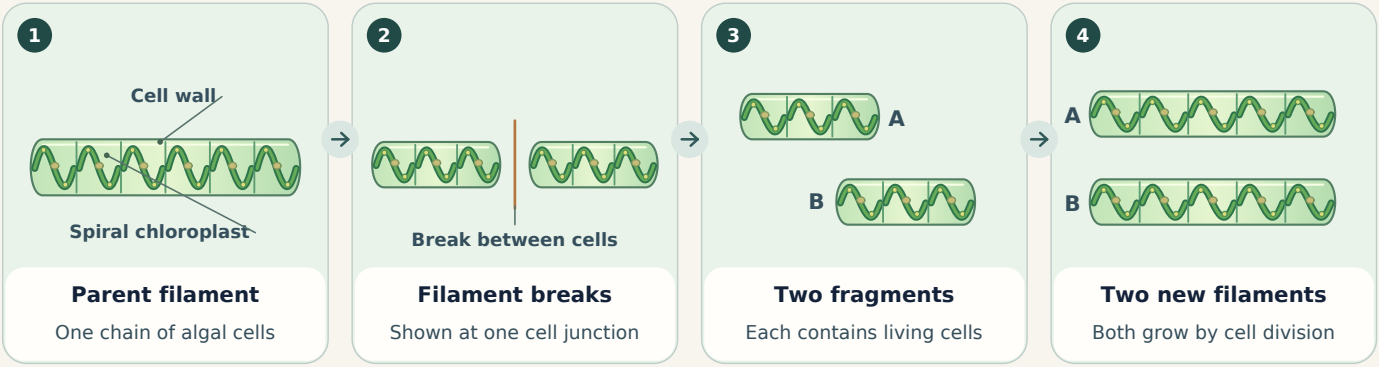
BUDDING



Repeated cell division in the parent's body wall forms the bud. As the bud matures, its connection to the parent narrows. The young Hydra separates and can feed on its own.

Spirogyra

FRAGMENTATION



Spirogyra is a green alga with ribbon-like chloroplasts that coil inside its cells. When its filament breaks into fragments, each viable piece can grow into an independent filament under favourable conditions.

TRACE THE PROCESS

At which stage does the Hydra bud start to look like the parent? Why does a Spirogyra fragment already contain several cells?

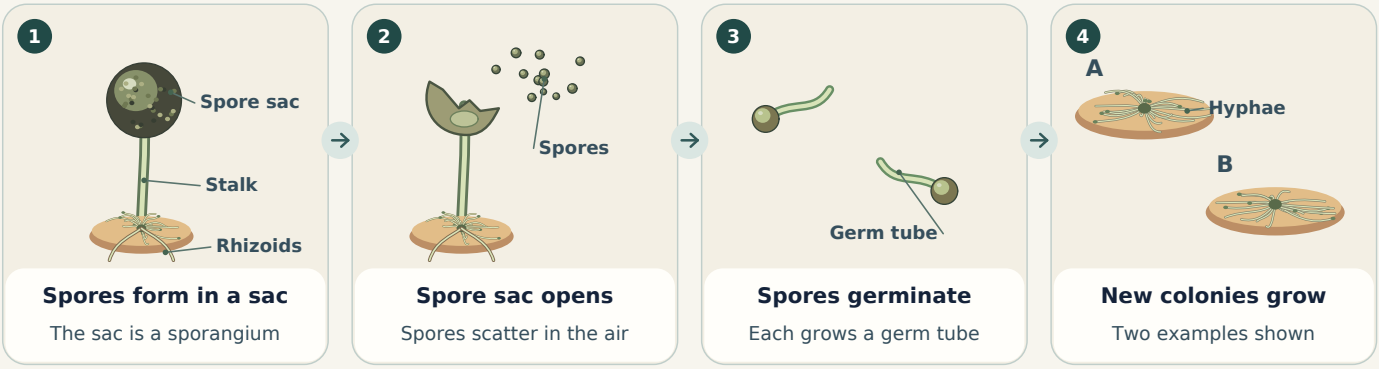
ASEXUAL REPRODUCTION

Spore Formation in Fungi

Air currents carry tiny, often thick-walled asexual fungal spores to new surfaces. A spore can germinate and grow into a colony where moisture, nutrients and temperature are suitable.

Rhizopus

BREAD MOULD



Rhizopus. Thread-like hyphae form a network on bread. Some grow upright as stalks and bear sporangia. When a mature sporangium breaks open, it releases the spores formed inside.

Aspergillus

CONIDIA



Aspergillus. The stalk ends in a swollen tip called a vesicle. Spore-producing cells on this tip form exposed chains of conidia. In both moulds, the germ tube grows and branches into a network of hyphae called a mycelium.

Fungi and people. Refrigeration slows the growth of many moulds. Fungi recycle nutrients and provided the starting compounds for medicines such as penicillin and related antibiotics such as amoxicillin. Researchers also study fungi for pollutant removal and plastic breakdown.

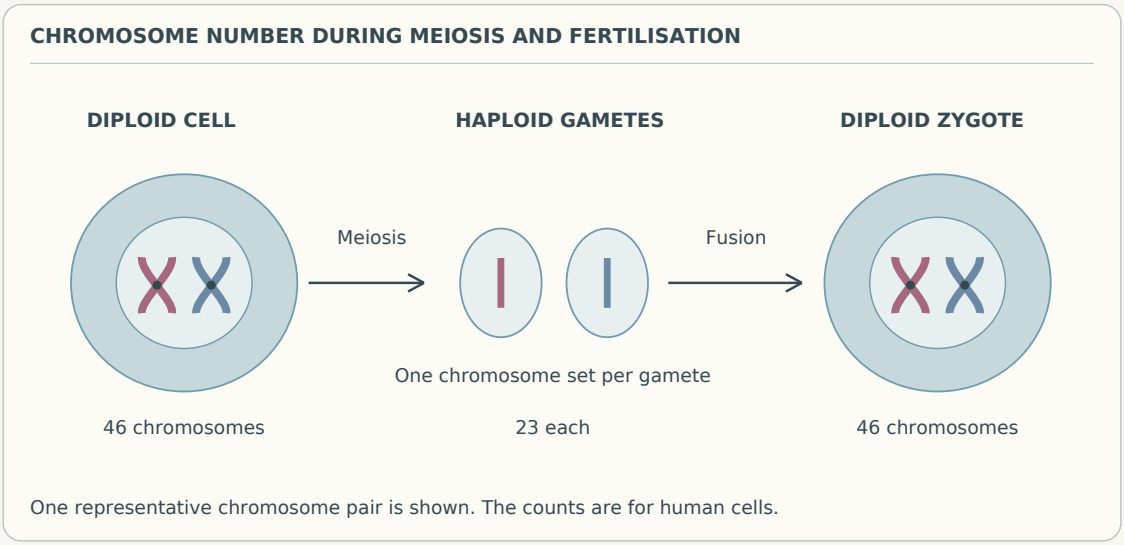
CHECKPOINT 1

Which grower's technique would keep the traits of a sweet mango? Why is budding asexual? Which mould produces spores inside a sac?

SEXUAL REPRODUCTION

Sexual Reproduction

Sexual reproduction depends on the fusion of a male gamete and a female gamete. In animals these are usually called **sperm** and **egg**. The zygote formed at fertilisation can divide repeatedly and develop into an embryo. If gametes contained two full sets of chromosomes, the number of chromosome sets would double each time gametes fused. **Meiosis** prevents this by reducing the number of chromosome sets.



Meiosis and Genetic Variation

Chromosomes are DNA-containing structures in the nucleus. Most human body cells have 46 chromosomes arranged in 23 pairs. A cell with two sets of chromosomes is **diploid**. A cell with one set is **haploid**. In animals, meiosis is part of gamete formation. Each human sperm or egg ordinarily carries 23 chromosomes. When they fuse, the zygote again has 46.

Why siblings differ

During meiosis, the two chromosomes of each pair separate. Different pairs assort independently, and paired chromosomes can exchange corresponding pieces of DNA. Fertilisation adds another source of variation because the sperm and egg that meet are only one possible combination. As a result, siblings usually inherit different combinations of genes. Identical twins are an exception because they develop from the same original zygote.

Gametes in flowering plants

In flowering plants, meiosis produces spores that develop into gamete-producing structures. A pollen grain carries or develops the male gametes. The ovule contains the egg. The next section shows how a flower brings the gametes together.

CHECKPOINT 2

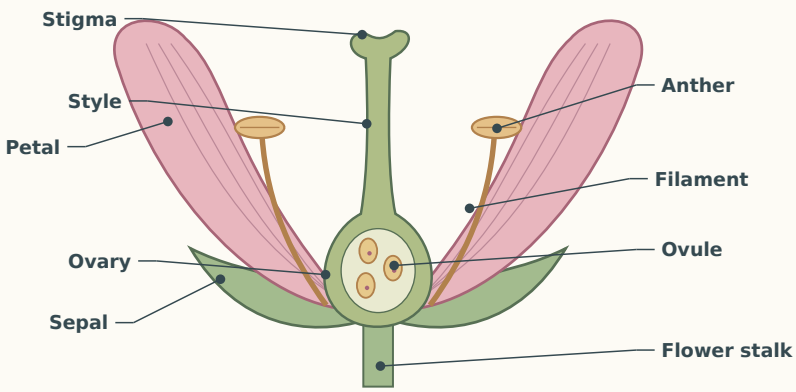
A species has 18 chromosomes in each body cell. Work out the chromosome number in a gamete and in a zygote.

FLOWERING PLANTS

Sexual Reproduction in Flowering Plants

Flowering plants are called **angiosperms**. A flower is their sexual reproductive structure. A complete flower has four sets of organs arranged in rings called **whorls**. From the outside to the centre, these are sepals, petals, stamens and the pistil. A longitudinal section is a cut along the flower’s length that reveals how these parts fit together.

LONGITUDINAL SECTION OF A COMPLETE FLOWER



Generalised flower. The section exposes the ovules inside the ovary.

The outer two whorls

Sepals are often green and protect the flower while it is still a bud. **Petals** often provide colour, scent or patterns that help pollinators locate the flower. Features differ among plants. Many flowers pollinated by wind have small or inconspicuous petals.

Stamen: the male reproductive part

A stamen consists of a **filament**, which is a slender stalk, and an **anther** at its tip. Pollen grains develop inside the anther. When the anther matures, it opens and releases pollen. In the diagram, the filaments hold the anthers away from the flower base, where moving air or visitors can reach them.

Pistil: the female reproductive part

The **stigma** is the receptive tip on which compatible pollen can land and germinate. The **style** connects the stigma to the **ovary**, the enlarged base of the pistil. The ovary contains one or more **ovules**. Each ovule contains the structure in which an egg cell develops. The pollen grain must reach the stigma before it can send a tube towards an ovule.

LABEL IT

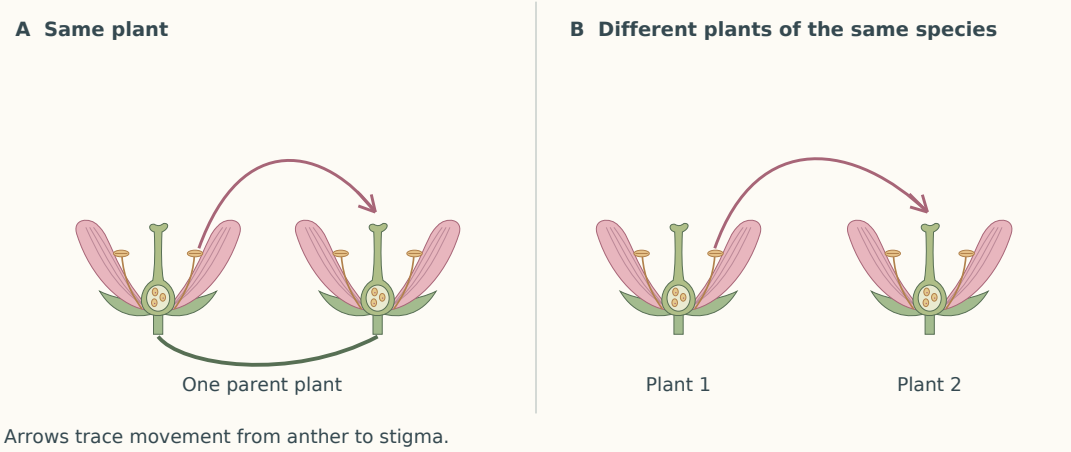
Cover the diagram labels and identify the four whorls from the outside inward. Then name the three regions of the pistil.

FLOWERING PLANTS

Pollination

Pollination is the transfer of pollen from an anther to a stigma. Once pollen reaches a compatible stigma, it may germinate and grow a pollen tube. Pollination is therefore a transport step. **Fertilisation** happens later, when a male gamete reaches and fuses with an egg inside the ovule.

SELF-POLLINATION AND CROSS-POLLINATION



Arrows trace movement from anther to stigma.

Self-pollination

Pollen moves to the stigma of the same flower or to another flower on the same plant. A plant that can self-pollinate can sometimes set seed even when a suitable partner or pollinator is scarce. Self-pollination can still produce different combinations of the parent's alleles, but it brings in no genes from another individual. Repeated selfing generally reduces genetic diversity in a population.

Cross-pollination

Pollen travels to a flower on a different individual of the same species. If fertilisation follows, offspring can inherit genetic material from two different plants. This often maintains more variation in a population. Cross-pollination depends on pollen reaching a compatible plant at the right stage.

Use the arrangement as evidence

In tomato, the stamens surround the stigma, making transfer within the same flower possible. Wheat commonly self-pollinates, often before the flower opens fully. In papaya plants that bear male and female flowers on separate individuals, pollen must move between plants. Maize is a familiar example of wind-assisted cross-pollination, although self-pollination can also occur.

CLASSIFY IT

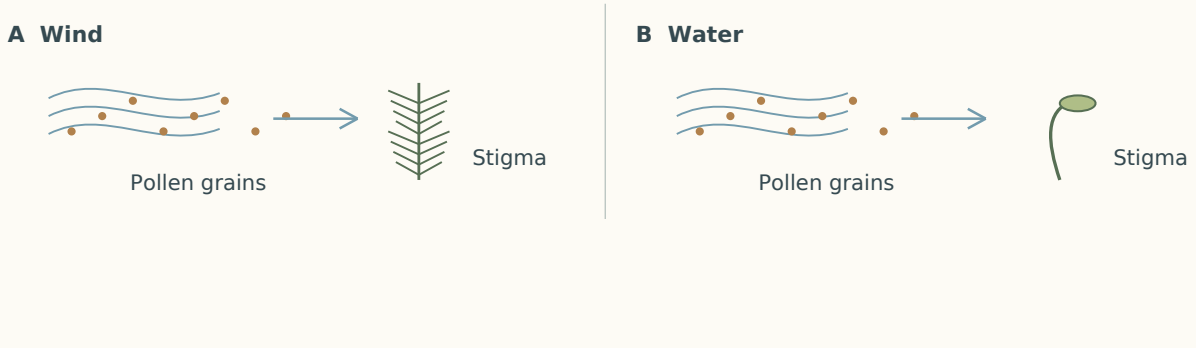
Pollen from one flower lands on a second flower on the same plant. Is the transfer self-pollination or cross-pollination?

FLOWERING PLANTS

Pollination Strategies and Reproductive Success

Pollen may be transported by wind, water or animals. The traits of a flower often match its usual means of transfer. A feathery stigma can catch airborne pollen, while a nectar-producing flower may encourage repeated visits by animals.

POLLINATION BY WIND AND WATER



Pollen is carried through air or water towards a receptive stigma.

Wind Pollination

Maize, wheat and rice are grasses whose pollen can be carried by air. Wind-transported pollen is generally light and produced in abundance because many grains miss a compatible stigma. Long, feathery stigmas in maize intercept airborne pollen. Wheat and rice commonly self-pollinate, so wind transport and the usual type of pollination are separate questions.

Water Pollination

In some aquatic plants, water moves pollen towards a receptive stigma. In *Vallisneria*, male flowers can detach and float to the surface, where female flowers receive pollen. Water-mediated transfer also occurs in *Hydrilla*. Many other aquatic plants use insects or wind, just as land plants do.

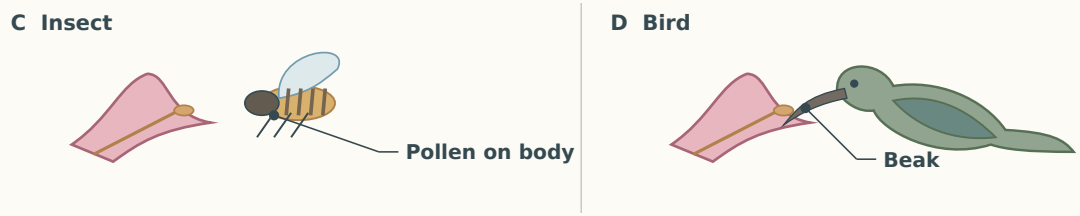
PREDICT THE AGENT

An exposed flower releases large quantities of light pollen and has a feathery stigma. Which means of transfer best fits those traits?

FLOWERING PLANTS

Insect and Bird Pollination

POLLINATION BY INSECTS AND BIRDS



A visitor can carry pollen when it moves from flower to flower.

Insect Pollination

Flowers such as sunflower, Hibiscus and marigold can attract insects through colour, scent or nectar. Pollen often sticks to a bee or butterfly and brushes onto a stigma when the animal visits another flower. Pale, strongly scented flowers that open at night can attract nocturnal insects such as moths.

Bird Pollination

Sunbirds and Indian white-eyes may carry pollen while feeding on nectar. Coral tree and some Hibiscus flowers can be visited by birds. In animal pollination, the route can be more targeted than wind dispersal. Pollen and seed counts still depend on species and on whether one flower or a cluster is being counted.

Compare transfer efficiency

The source gives an illustrative comparison: wind-pollinated grasses release about 500,000–1,000,000 pollen grains and form about 50–200 seeds, while the insect-pollinated example lists 20,000–40,000 pollen grains and 800–1,000 seeds. These counts are examples rather than fixed ratios. A sunflower head contains many small flowers, so a count for a head cannot be compared directly with a count for one flower. The biological point is that directed visits can deliver a greater fraction of pollen to receptive stigmas.

FOLLOW A VISIT

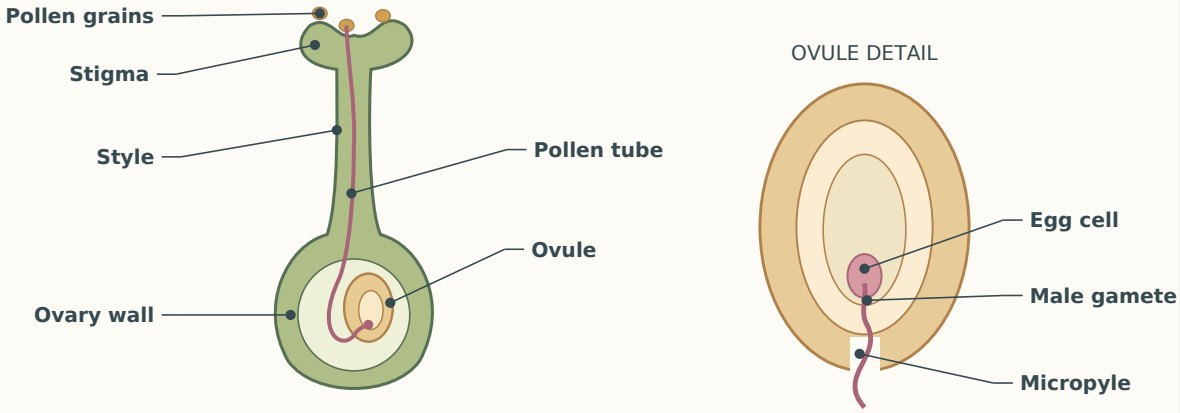
Describe how an insect can pick up pollen and deposit it at a second flower. Name both flower structures involved.

FLOWERING PLANTS

Fertilisation and Seed Formation

When a compatible pollen grain lands on a stigma, it absorbs moisture and germinates. A **pollen tube** grows through the style towards the ovary. The tube carries male gametes and usually enters an ovule through a small opening called the **micropyle**. One male gamete fuses with the egg. This is **fertilisation**, and it produces a zygote.

POLLEN TUBE GROWTH AND ENTRY INTO AN OVULE



Enlarged schematic. The pollen tube enters the ovule through the micropyle.

From ovule to seed

The zygote divides by mitosis to form an embryo. The ovule develops into a **seed**, which contains the embryo and a protective seed coat. In flowering plants, a second fusion normally helps form nutritive tissue called **endosperm**. A seed can remain dormant until conditions support growth.

From ovary to fruit

The ovary usually enlarges and becomes a **fruit**. The ovary wall contributes to the fruit wall, and seeds develop inside it. A fruit may protect the seeds and help disperse them. The pollen tube diagram shows the stage before the ovary and ovules take on these new roles.

Dispersal and germination

Wind, water and animals move seeds away from the parent. Dandelion fruits have a feathery **pappus**. Madar seeds have silky hairs. Both structures help the propagule travel in air. A dispersed seed can germinate if it receives suitable moisture, oxygen and temperature. Its first root anchors the seedling and begins to absorb water.

SEQUENCE IT

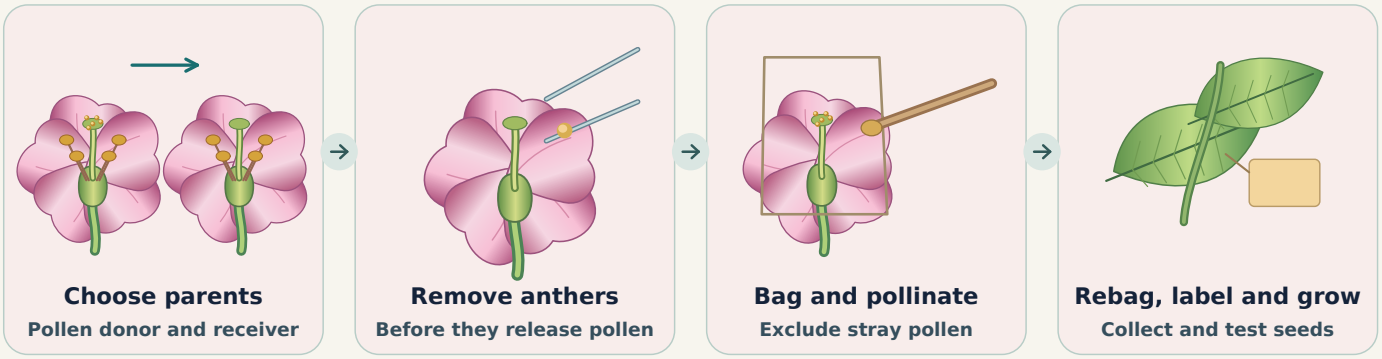
Put these in order: fertilisation, pollination, seed formation, pollen tube growth.

FLOWERING PLANTS

Plant Breeding and Its Applications

Plant breeders select parents with useful traits, then examine the offspring for combinations they want to keep. For example, one variety may have good fruit quality while another tolerates a particular disease. A controlled cross produces offspring with new combinations. Breeders select promising plants and test how well the traits persist and perform.

CONTROLLED CROSS IN A FLOWER



When the stigma is receptive, use only pollen from the chosen donor.

Breeding mixes inherited variants | **Engineering targets selected DNA**

Artificial hybridisation

Choose the plant that will receive the pollen and the plant that will supply it. If the receiving flower has stamens, remove its immature anthers before they release pollen. This step is called **emasculation**. Cover the flower with a bag to keep stray pollen away. When the stigma is receptive, apply the selected pollen, cover it again and label the cross. Later, collect and grow the seeds.

Genetic engineering

Breeding crosses combine many inherited differences at once. **Genetic engineering** can introduce or alter specific genetic information to obtain a selected trait. Pest resistance, drought tolerance and yield-related traits are examples of goals researchers may investigate. A trait still has to be tested under the conditions in which the crop will grow.

Scientist in focus: P. Maheshwari

Panchanan Maheshwari was a pioneer of Indian plant embryology. He advanced experimental approaches to studying fertilisation and embryo development in flowering plants, including work with plant material outside the plant. These methods helped researchers investigate crosses and early development under controlled conditions.

EXPLAIN THE METHOD

Why does a breeder remove immature anthers and cover the receiving flower before adding selected pollen?

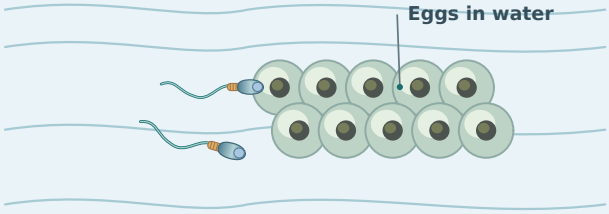
ANIMALS

Sexual Reproduction in Animals

In animals, sperm and egg must meet for sexual reproduction to occur. The site of their fusion gives us two categories: **external fertilisation** and **internal fertilisation**. The difference is where gametes unite. Whether the young later hatch from a laid egg or are born alive is a separate question.

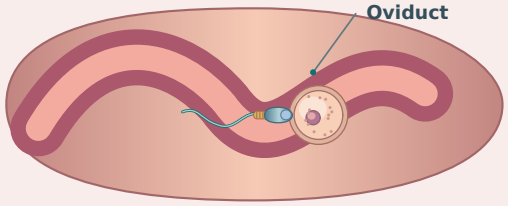
WHERE DO THE GAMETES MEET?

A EXTERNAL / FROG EXAMPLE



Sperm and eggs unite in water

B INTERNAL / BIRD EXAMPLE



Gametes unite inside the oviduct

The place of fertilisation and the place of development are separate ideas.

External Fertilisation

In many fish and amphibians such as frogs, females release eggs into water and males release sperm nearby. Water allows sperm to swim to the eggs. Timing matters because currents can separate gametes and sperm may remain able to fertilise for only a limited period. Predators can also eat eggs or newly hatched young. Releasing many gametes increases the chance that some eggs will be fertilised and some offspring will survive.

Internal Fertilisation

In reptiles, birds and mammals, sperm are transferred into the female reproductive tract. Gametes meet there and are sheltered from many external hazards. In birds and many reptiles, the fertilised egg develops within a laid egg. In placental mammals, the embryo develops inside the uterus. Internal fertilisation therefore describes the site of fusion, even when later development takes place outside the body.

What to compare

External fertilisation is common in water, although there are fish that use internal fertilisation. Internal fertilisation can occur on land or in water. Many internally fertilising species produce fewer eggs or young per reproductive event, but the pattern depends on the species and the care it gives its offspring.

APPLY IT

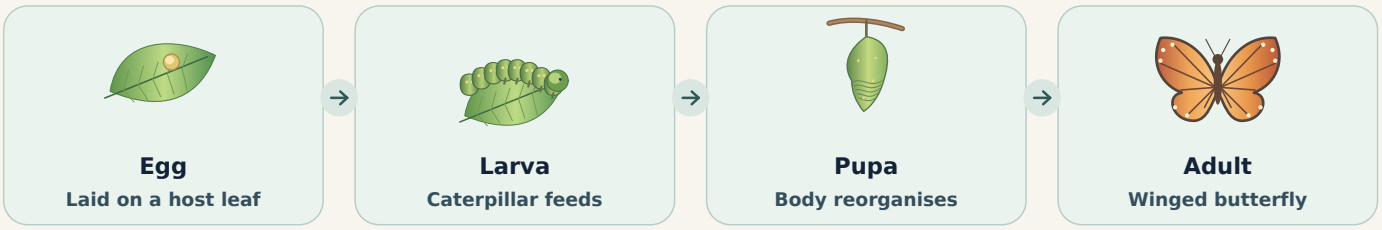
A bird lays an egg that hatches outside its body. Why is the bird's fertilisation still called internal?

ANIMALS

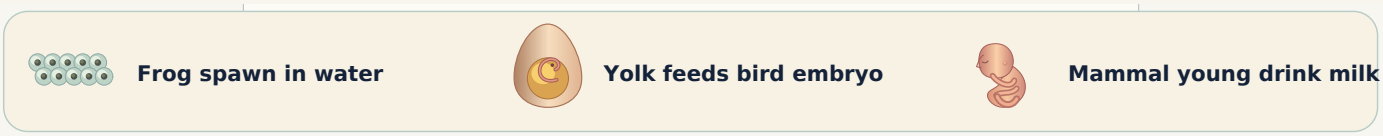
Variations in Reproduction in Animals

Animal reproductive strategies vary in the number of eggs, the food available to embryos and the level of parental care. Some species produce many young, few of which survive. Others invest more resources in a smaller number. These patterns can be understood by considering what happens between fertilisation and independent life.

BUTTERFLY / COMPLETE METAMORPHOSIS



A larva feeds and grows. An adult emerges after the pupal stage.



Eggs, larvae and metamorphosis

Many fish, amphibians and insects hatch at a **larval** stage. The egg's yolk supports early development. After hatching, the larva takes in food and grows. In a butterfly, the larva is a caterpillar. After feeding and growing, it forms a **pupa**, or chrysalis, and its body reorganises during **metamorphosis**. The adult butterfly emerges with wings and is able to reproduce.

Larger eggs and parental care

Many reptiles and birds produce relatively few eggs with enough yolk to support the embryo until a recognisable young animal hatches. Some reptiles guard their eggs, and many birds incubate their eggs and feed their chicks. There are exceptions to these patterns. Clutch size and survival vary with species and habitat. As illustrative ranges, the source lists hundreds to thousands of eggs for fish, 5,000–50,000 for some frogs, 2–20 for lizards and 1–15 for birds. Even within a group, a smaller clutch can come with more care per egg.

Mammals

In placental mammals, the embryo grows inside the mother's body. A **placenta** supports exchange of nutrients, oxygen and waste. After birth, the young are fed milk. Human infants depend on caregivers for a long time. That care supports survival and allows extended brain development, learning and social development.

CHECKPOINT 3

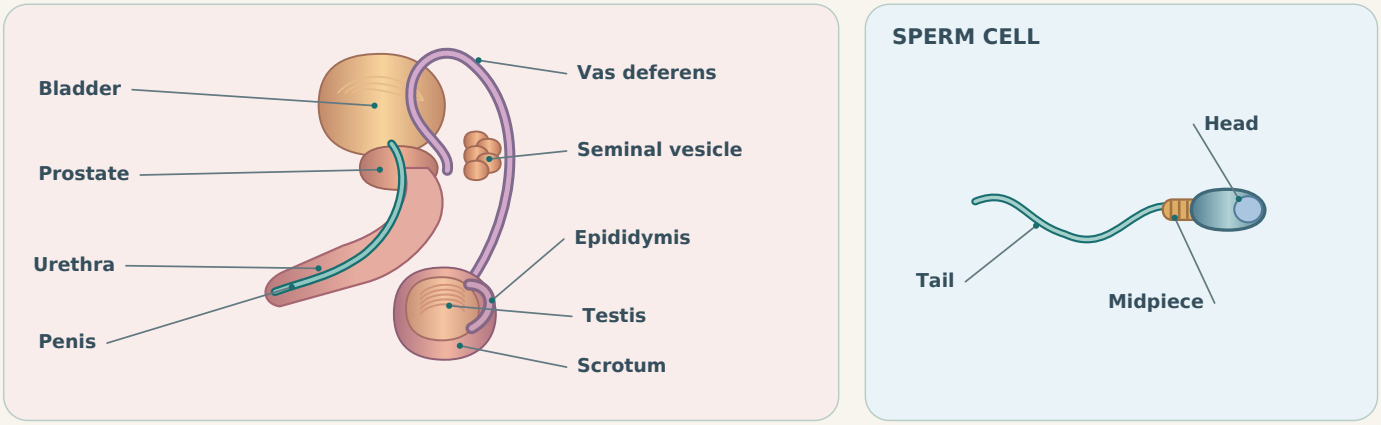
How do wind-pollinated flowers improve pollen capture? What does a caterpillar do before pupating? Give one advantage of internal fertilisation.

HUMAN REPRODUCTION

Reproduction in Human Beings

During puberty, hormonal changes bring about physical growth and maturation of the reproductive organs. Testes begin producing sperm and ovaries begin releasing egg cells as ovulatory cycles become established. The timing varies among individuals. Other changes can include a growth spurt, body hair and changes in voice or body shape.

MALE REPRODUCTIVE SYSTEM / SIDE VIEW



SPERM ROUTE Testis → Epididymis → Vas deferens → Urethra

Educational cutaway and enlarged sperm. The structures use different scales.

Male Reproductive System

The two **testes** produce sperm and the hormone testosterone. The testes lie in the **scrotum**, which helps maintain a temperature slightly lower than core body temperature. Newly formed sperm pass into the **epididymis**, a coiled duct where they mature and are stored.

The pathway of sperm

The **vas deferens** carries sperm from the epididymis towards the urethra. The **seminal vesicles** and **prostate gland** add fluids that help nourish and transport the sperm. Sperm mixed with these fluids form **semen**. Semen leaves the body through the **urethra**, which runs through the penis. The urethra also carries urine at other times.

Structure of a sperm

The head contains the nucleus and carries genetic information. The midpiece provides energy for movement, and the long tail, or flagellum, helps propel the cell. A typical sperm cell contains 23 chromosomes. Its small size and motility suit its role in reaching an egg.

TRACE IT

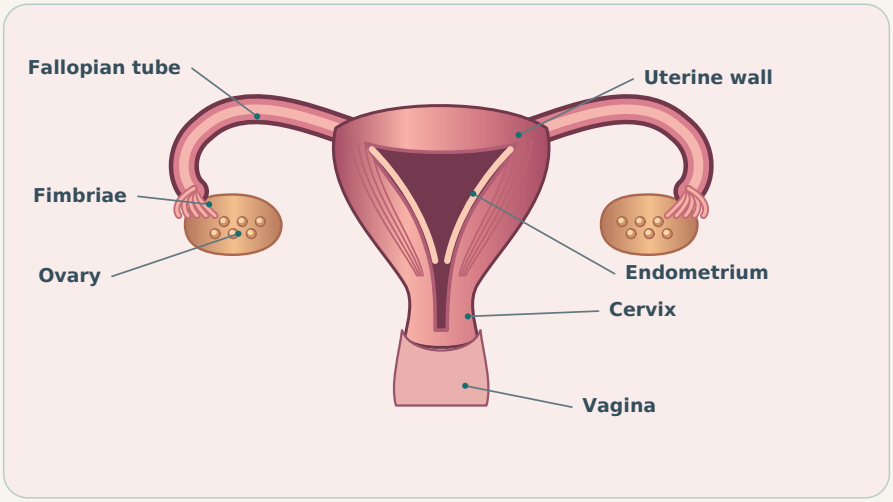
Follow the sperm pathway from the testes to the urethra. At which stage do gland secretions become part of semen?

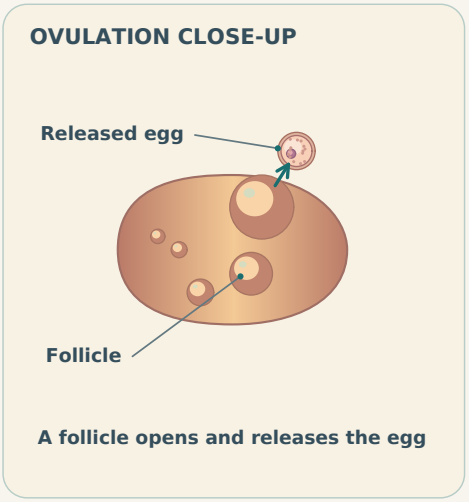
HUMAN REPRODUCTION

Female Reproductive System

The female reproductive system releases egg cells, provides the usual site of fertilisation and supports development if pregnancy begins. The organs involved include the ovaries, Fallopian tubes, uterus, cervix and vagina. The schematic shows them from the front so the two sides are easy to compare.

FEMALE REPRODUCTIVE SYSTEM / FRONT CUTAWAY





The ovaries sit beside the open, fringed ends of the tubes.

Ovaries and Fallopian tubes

The two **ovaries** contain egg cells in follicles and produce hormones including oestrogen and progesterone. At ovulation a mature egg is released. Finger-like structures called **fimbriae** help guide it towards the nearby **Fallopian tube**, also called the oviduct. The egg travels along the tube towards the uterus. The tube is the usual site of fertilisation.

Uterus and its lining

The **uterus** is a thick-walled muscular organ. Its inner lining, the **endometrium**, changes over the menstrual cycle. If pregnancy begins, the embryo implants in this lining and develops within the uterus. The muscular wall also contracts during childbirth.

Cervix and vagina

The **cervix** is the narrow lower part of the uterus. Its canal connects the uterus to the **vagina**, a muscular passage. The vagina receives semen during intercourse, allows menstrual fluid to leave the body and serves as the birth canal during vaginal delivery.

COMPARE LOCATIONS

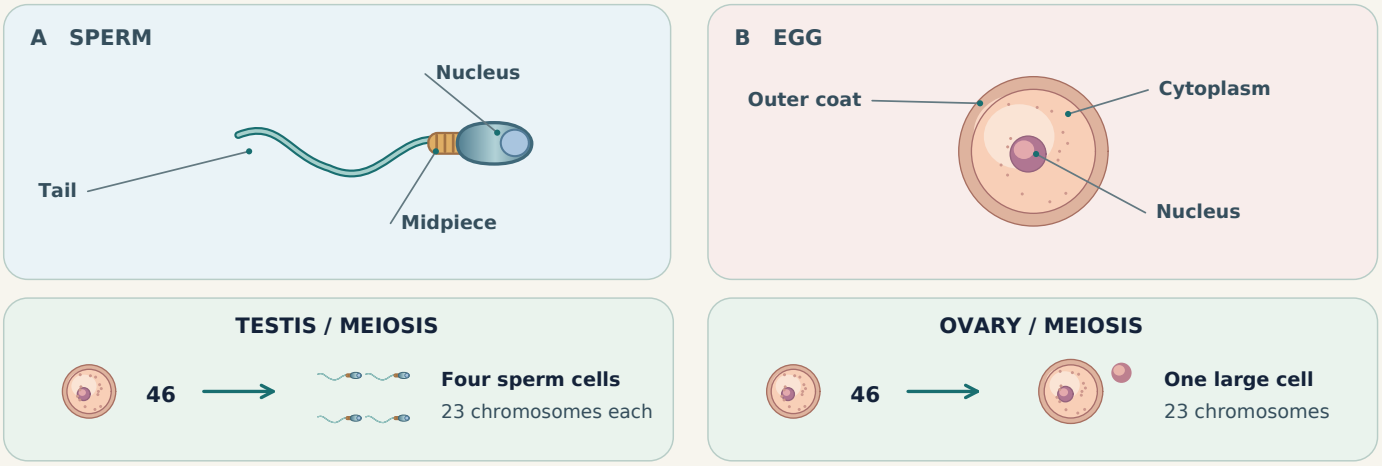
Which organ releases the egg? Where does fertilisation usually occur? Where does implantation occur?

HUMAN REPRODUCTION

Gametogenesis

Gametogenesis is the formation of gametes. In humans it occurs in the testes and ovaries and involves meiosis. Meiosis reduces the chromosome number from 46 in a diploid starting cell to 23 in a haploid gamete. Fertilisation brings together two haploid sets, restoring the diploid number of 46.

GAMETES / STRUCTURE AND ORIGIN



The drawings use different scales. Human gametes carry 23 chromosomes each.

Sperm formation

Sperm production is called **spermatogenesis**. From puberty, cells in the testes divide and develop into sperm. Meiosis produces cells with one chromosome set. As they mature, they become streamlined and develop a tail. Millions can be produced, and only a small proportion of the sperm released will reach an egg.

Egg formation

Egg formation is called **oogenesis**. It begins before birth. After puberty, an immature egg in a follicle can resume development during an ovarian cycle. Usually one egg is released in an ovulatory cycle. The egg is much larger than a sperm and retains more cytoplasm and stored materials that support the first stages of development.

Compare the gametes

A sperm actively swims with a tail. An egg has no tail and is transported by movements in the Fallopian tube. The gametes differ in size, shape, motility, the amount of cytoplasm and the number typically produced or released. Both carry one chromosome set and each supplies part of the zygote's genetic information.

WORK IT OUT

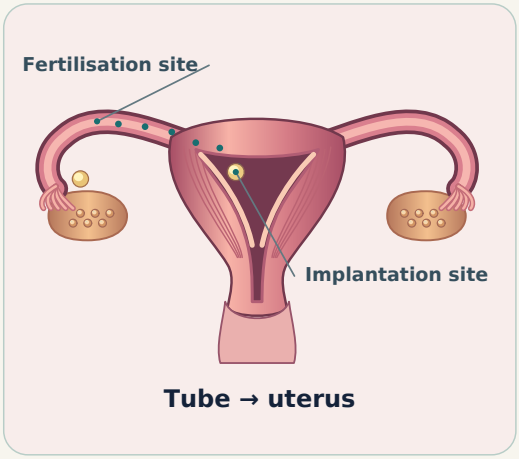
Why does a zygote have 46 chromosomes even though an egg and a sperm each have 23?

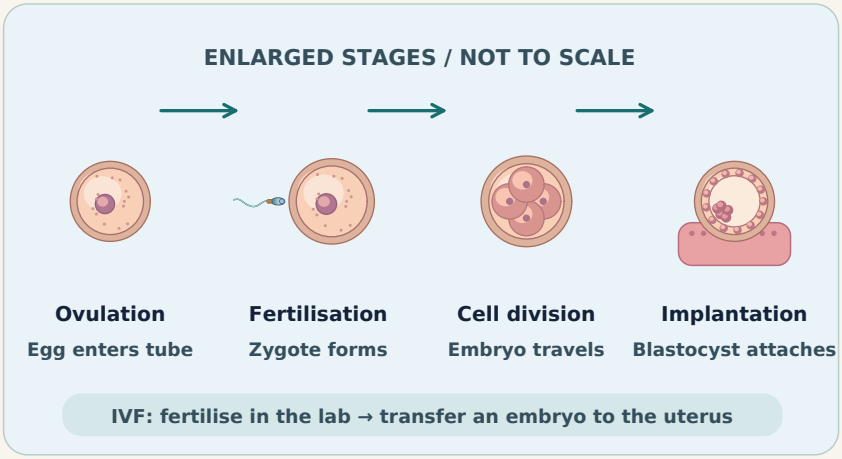
HUMAN REPRODUCTION

Fertilisation and Implantation

At **ovulation**, an egg leaves an ovary and enters a Fallopian tube. Sperm deposited in the vagina can travel through the cervix and uterus into a tube. Many are lost or fail to complete the journey. If sperm and egg meet, normally one sperm fuses with the egg. This event is **fertilisation**, and the resulting single cell is a **zygote**.

FROM OVULATION TO IMPLANTATION





Typical fertilisation is in a Fallopian tube. Implantation occurs in the uterine lining.

The first divisions

The zygote divides repeatedly by mitosis while moving towards the uterus. At first the cells become more numerous without the whole structure growing much in size. The developing embryo later forms a hollow stage called a **blastocyst**. It reaches the uterus and comes into contact with the thickened endometrium.

Implantation

Several days after fertilisation, the blastocyst attaches to the uterine lining and begins embedding in it. This is **implantation**. Follow the sequence on the diagram. Fertilisation is shown in the Fallopian tube. Implantation is shown at the uterine lining, where the embryo will continue to develop.

Scientist in focus: Subhash Mukhopadhyay

In **in vitro fertilisation** (IVF), an egg is fertilised outside the body in a laboratory. An embryo may then be transferred to the uterus, where implantation may occur. In 1978, Subhash Mukhopadhyay and his team helped achieve India's first IVF birth, Kanupriya Agarwal, known as Durga.

PUT IN ORDER

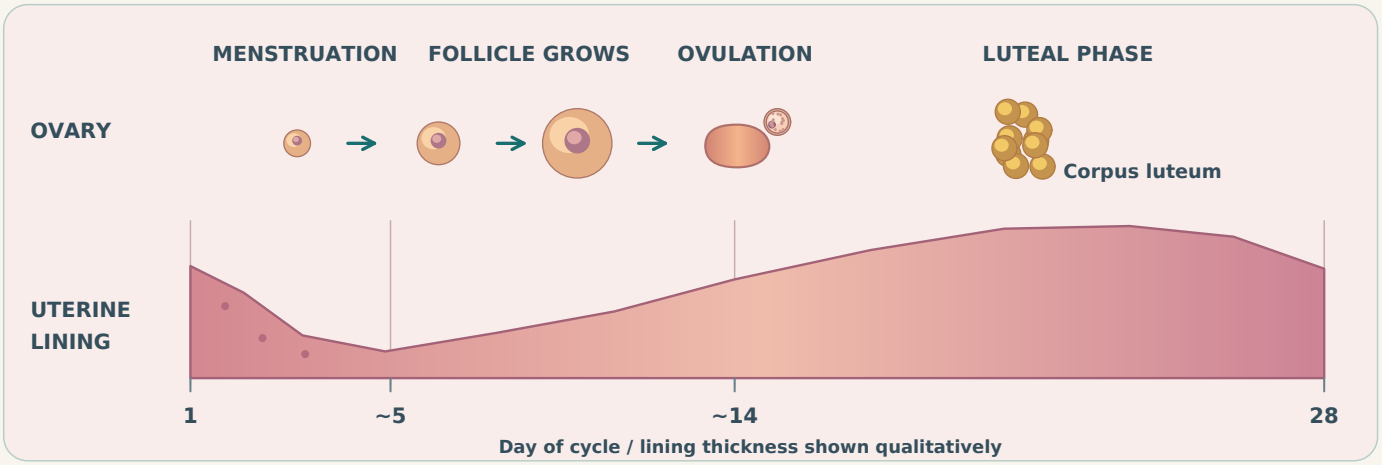
Start at ovulation and put fertilisation, early mitotic divisions and implantation in order. Name the organ for each of the two major events.

HUMAN REPRODUCTION

Menstrual Cycle

The **menstrual cycle** is a repeating pattern of changes in the ovaries and uterus. Day 1 is the first day of menstrual bleeding. A cycle is counted from day 1 of one period to day 1 of the next. A 28-day cycle is a useful model. It illustrates the sequence, while real cycle lengths and the timing of ovulation vary.

OVARY AND UTERUS / A MODEL 28-DAY CYCLE



Day 14 is an example. Real cycle length and ovulation timing vary.

Menstruation and the follicular phase

When pregnancy has not begun, levels of hormones that support the uterine lining fall. The lining sheds, and blood and tissue leave through the vagina. In a 28-day example, bleeding is often shown on days 1–5. The **follicular phase** starts on day 1 and continues until ovulation. A follicle develops in the ovary, and rising oestrogen helps the uterine lining rebuild.

Ovulation and the luteal phase

Ovulation is shown around day 14 in the model. The follicle releases an egg. In the **luteal phase**, the tissue left by the follicle produces hormones, including progesterone, that help maintain the lining. If pregnancy begins, hormonal signals maintain the lining and normal cycles pause. Otherwise, hormone levels fall and a new period begins.

Variation across life

The first period is called **menarche** and usually occurs during puberty. In adolescence, cycles often vary. Many are about 21–45 days, and bleeding commonly lasts 2–7 days. Menstruation ends at **menopause**, usually around age 50. Day 14 is a teaching example, and a calendar model alone cannot establish the exact day of ovulation.

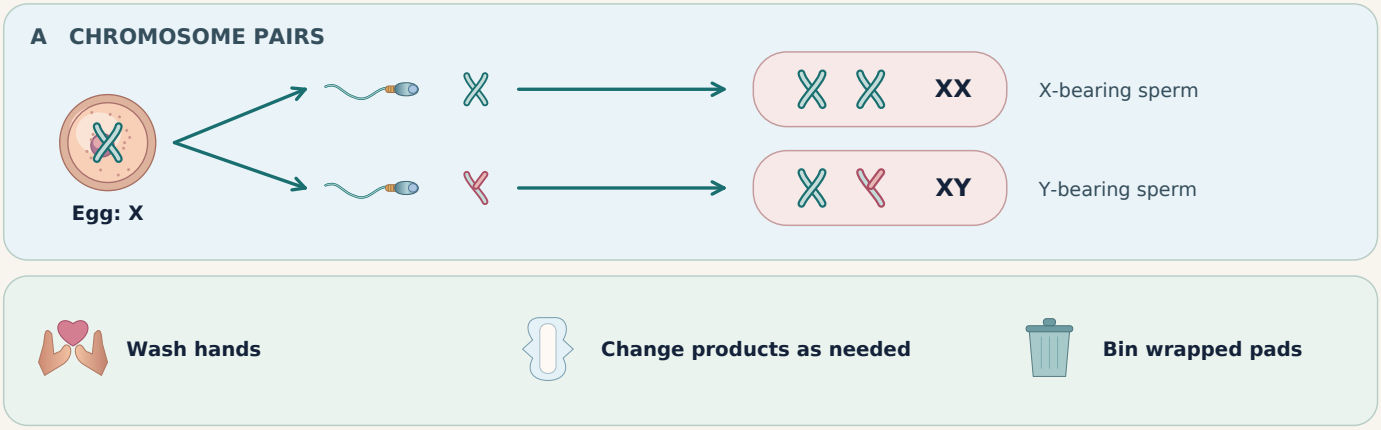
CALENDAR EXAMPLE

If day 1 is 5 March and the cycle lasts 28 days, the next day 1 would be 2 April.

HUMAN REPRODUCTION

Curiosity Corner and Menstrual Hygiene

XX/XY MODEL AND MENSTRUAL HYGIENE



The two chromosome branches represent separate possible fertilisation events.

Curiosity Corner: Sex chromosomes

In the common XX/XY pattern, most body cells have a pair of sex chromosomes. A typical egg carries one X chromosome. A typical sperm carries either an X or a Y. When an X-bearing sperm fuses with the egg, the resulting pair is XX. When a Y-bearing sperm fuses with it, the pair is XY. XX is commonly associated with female development and XY with male development. Variations in chromosomes and in biological sex development also occur.

Which sperm reaches and fertilises the egg is a chance event. An egg supplies an X chromosome in either branch of the model. The model explains how the two usual chromosome combinations can arise at fertilisation.

Menstrual Hygiene

Menstruation is a normal bodily process. Sanitary pads, tampons and menstrual cups are products that can be used to manage menstrual flow. The choice depends on comfort, access and suitability. Use clean hands before and after changing a product and follow its instructions. Change pads every few hours or sooner if soaked, and use clean water for external washing.

Used disposable pads can be wrapped and put in an appropriate bin. Flushing them can block toilets. A reusable product needs to be cleaned as directed. Access to clean water, suitable products and a private place to change helps students manage periods comfortably at school. Family members and teachers can help reduce stigma by using accurate language and offering practical support.

READ THE FIGURE

In the simplified XX/XY model, which chromosome can differ between the sperm in the two branches?

HUMAN REPRODUCTION

Pregnancy and Childbirth

Pregnancy lasts about nine months and is commonly divided into three **trimesters**. Early development begins after fertilisation, and implantation places the embryo within the uterine lining. The **placenta** develops at the maternal–foetal interface. Oxygen and nutrients can pass to the developing baby, and wastes can pass in the opposite direction. The **umbilical cord** connects the developing baby to the placenta.

PREGNANCY / DEVELOPMENT AND EXCHANGE



Selected stages are shown at different scales. Maternal and foetal blood normally remain separate.

First trimester | Months 1–3

The major body plan begins to form. Structures such as the brain, heart and limbs start developing during the embryonic period. After eight weeks of development following fertilisation, the developing individual is called a **foetus**. The first image represents an early stage of development.

Second trimester | Months 4–6

Growth continues and body features become clearer. The foetus moves and its movements may become noticeable to the pregnant person. The organs continue to develop, and the placenta and umbilical cord continue to support exchange.

Third trimester | Months 7–9

The foetus gains weight and the lungs continue to mature. Many foetuses move into a head-down position near birth. Position can vary. The diagram shows one possible position near the end of pregnancy.

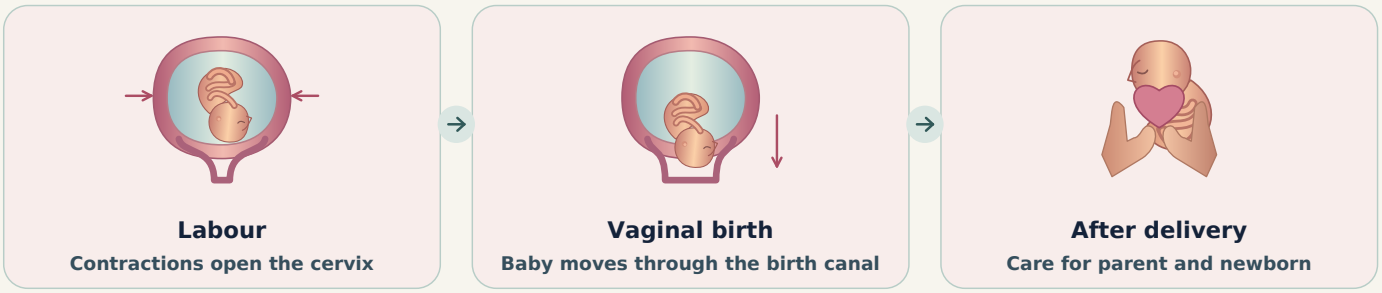
CONNECT IT

Name the exchange organ and the structure that connects it to the developing baby.

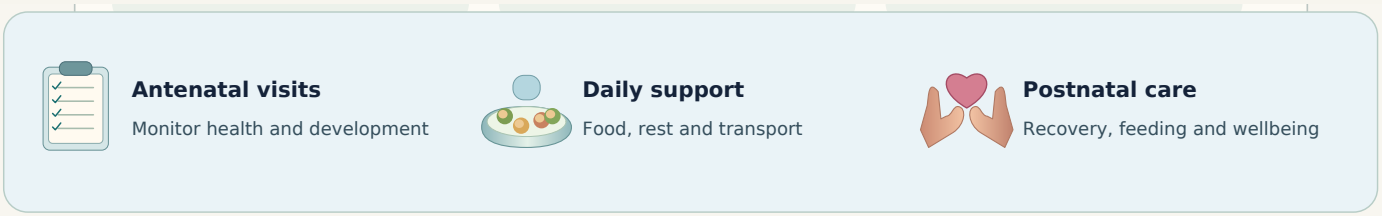
HUMAN REPRODUCTION

Childbirth and Care

CHILDBIRTH / FROM LABOUR TO EARLY CARE



A schematic of vaginal birth. A caesarean birth follows a different route.



Childbirth

During labour, contractions of the uterus help open the cervix. In vaginal birth, contractions help move the baby through the birth canal. A caesarean birth is a surgical delivery that may be used when needed. After birth, breast milk provides nourishment and protective antibodies.

Maternal health

Foetal development depends in part on the conditions in which pregnancy progresses. Antenatal care supports the health of the pregnant person and the developing baby. Regular check-ups give health workers opportunities to monitor development, discuss symptoms and provide support.

Nutrition, activity and rest

A balanced diet supplies proteins, vitamins and minerals that support health and growth. Suitable activity and adequate rest can help during pregnancy. A health professional can help assess the needs of an individual, including when extra nutrition, changes to activity or treatment are needed.

Avoiding harmful exposure

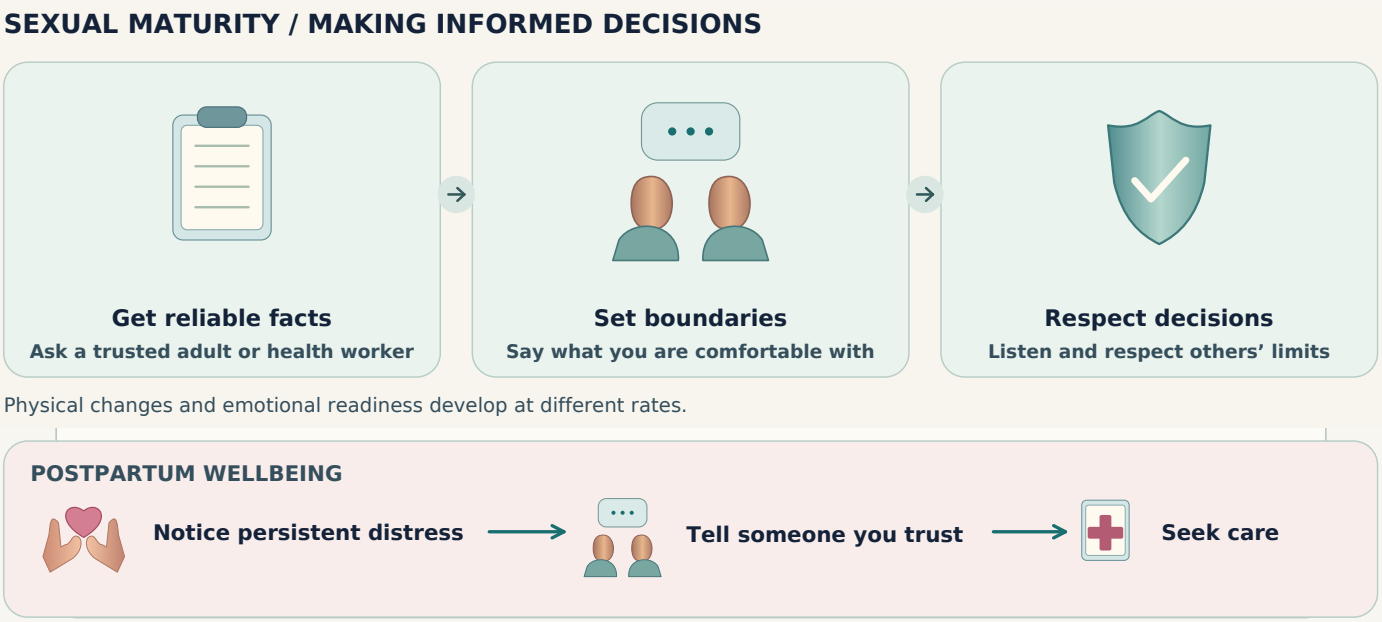
Tobacco and alcohol can harm development. A health professional can advise on medicines and other substances during pregnancy. Support from family and healthcare workers can help someone access appointments, rest, nutrition and emotional support.

DISCUSS

Give two ways practical support can help during pregnancy or after childbirth.

REPRODUCTIVE HEALTH

Postpartum Health and Sexual Maturity



Postpartum depression

After birth, some people experience persistent sadness, anxiety or severe exhaustion that interferes with daily life. **Postpartum depression** is a health condition that can be treated. Timely support from a doctor or nurse can help. Community workers such as ASHAs can help families find appropriate care.

Sexual Maturity and Responsibility

Sexual maturity is the development of the body's capacity to reproduce. Puberty brings physical changes, while emotional and social readiness develop over time. Responsible decisions involve respecting personal boundaries, seeking reliable information and recognising the importance of freely given consent. A trusted adult or health professional can help students find accurate answers about their bodies and health.

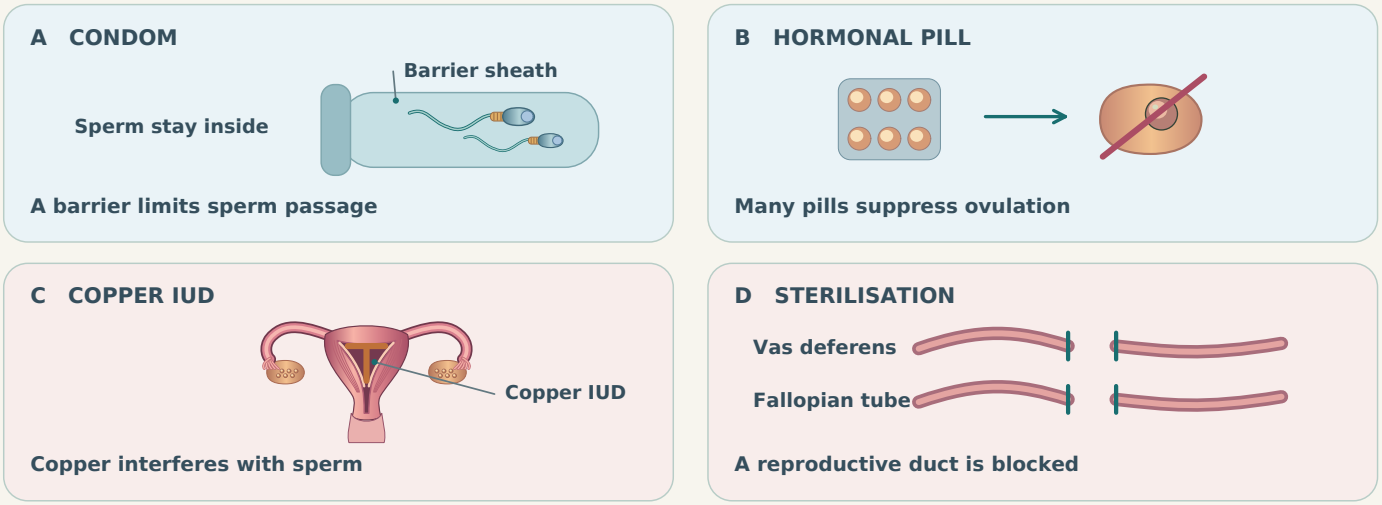
CHECK UNDERSTANDING

Give an example of an informed decision that supports reproductive health or respects a personal boundary.

REPRODUCTIVE HEALTH

Prevention of STIs and Unintended Pregnancy

FOUR METHODS / HOW THEY WORK



Condoms also lower the risk of many STIs. The other methods shown do not provide STI protection.

Sexually transmitted infections

STIs are infections that can spread through sexual contact. Examples include gonorrhoea, syphilis, herpes, genital warts caused by HPV, and HIV, which can progress to AIDS without treatment. Some STIs can be cured, and others can be managed with treatment. Correct and consistent condom use reduces the risk of many STIs and unintended pregnancy.

Barrier and hormonal methods

A **condom** acts as a physical barrier and helps prevent sperm from entering the reproductive tract. Many **hormonal pills** prevent ovulation and can have side effects. Hormonal methods do not provide a barrier against STIs.

Intrauterine devices and sterilisation

A clinician places an **IUD** such as a copper IUD inside the uterus. It provides long-term, reversible contraception. **Vasectomy** blocks the vas deferens, and **tubectomy** blocks the Fallopian tubes. Both procedures are intended to be permanent. IUDs and sterilisation do not protect against STIs.

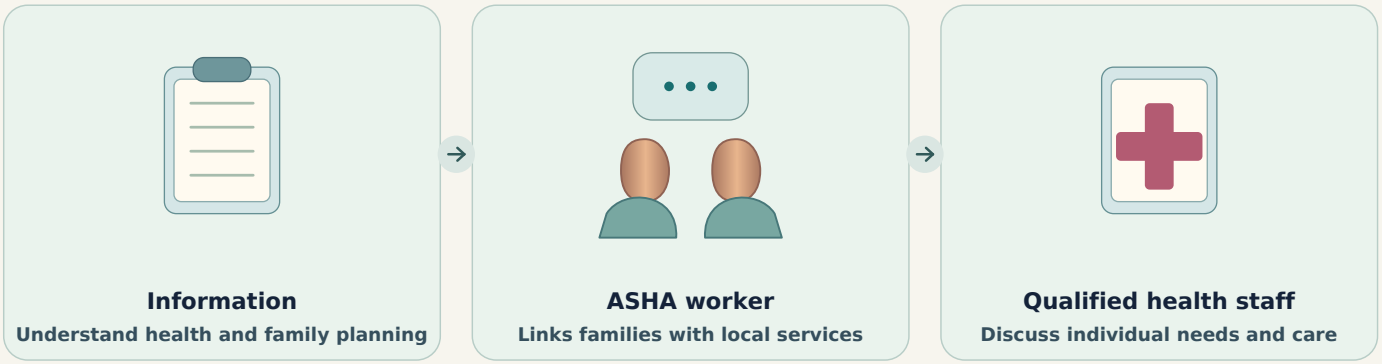
COMPARE METHODS

Which method reduces the risk of both unintended pregnancy and many STIs? Explain how its barrier helps.

REPRODUCTIVE HEALTH

Family Planning and Community Health

COMMUNITY REPRODUCTIVE HEALTH / A PATH TO CARE



Support can include information, referrals and individual clinical care.

REVERSIBLE
Examples include pills and IUDs

INTENDED TO BE PERMANENT
Example: sterilisation

Medical termination and sex selection

Abortion ends an established pregnancy and can involve medicines or a procedure. It is distinct from contraception, which helps prevent pregnancy. India's PCPNDT Act prohibits sex selection and the use of prenatal diagnostic techniques to determine foetal sex. Sex-selective practices can distort the population sex ratio.

ASHA workers and family planning in India

Accredited Social Health Activists (**ASHAs**) are community health workers who provide information about maternal health, safer delivery and family planning. Centchroman, also called Chhaya, was developed at CDRI in Lucknow. It is a non-hormonal oral contraceptive offered through India's public family planning programme.

Making sense of the choices

Some contraceptive methods are reversible and some are intended to be permanent. Methods also differ in how they work. A health professional can explain which options suit a person's health and circumstances. Reproductive health services also include guidance about menstrual health, pregnancy care and the prevention and treatment of infections.

CHECKPOINT 4

Why does the uterine lining thicken? What happens to menstruation during pregnancy? Name two contraception methods that do not protect against STIs.

REVISION

Mind Map and Key Distinctions

FOUR REPRODUCTION PATHWAYS

	**Asexual**	**One parent**	→	**Mitosis and growth**	→	**New individual**
	**Sexual**	**Meiosis**	→	**Gamete fusion**	→	**Zygote**
	**Flower**	**Pollination**	→	**Fertilisation**	→	**Seed and fruit**
	**Human**	**Ovulation**	→	**Fertilisation**	→	**Implantation**

Follow one row at a time. Explain what changes at each arrow.

Asexual reproduction

A single parent produces new individuals without gamete fusion. Mitosis maintains chromosome number as cells divide. In plants, roots, stems and leaves may produce new plants. In yeast and Hydra, buds grow from the parent. In fungi such as Rhizopus, spores are formed and dispersed.

Sexual reproduction

Meiosis reduces chromosome number and creates new genetic combinations. When two gametes fuse at fertilisation, the zygote regains two sets. Genetic variation arises during meiosis and through the combinations of gametes that meet.

Flowering plants

Pollination transfers pollen to a stigma. A compatible pollen grain grows a tube through the style. A male gamete fuses with an egg in the ovule. The zygote forms an embryo, the ovule develops into a seed and the ovary usually becomes a fruit.

Animals and humans

External and internal fertilisation describe the site where gametes meet. In humans, fertilisation normally takes place in a Fallopian tube. The embryo divides and later implants in the uterine lining. During pregnancy, the placenta and umbilical cord help support development.

Reproductive health

A 28-day menstrual cycle is a model. Menstrual hygiene includes clean hands and appropriate product use and disposal. Contraception can help prevent unintended pregnancy. Condoms also reduce the risk of many sexually transmitted infections.

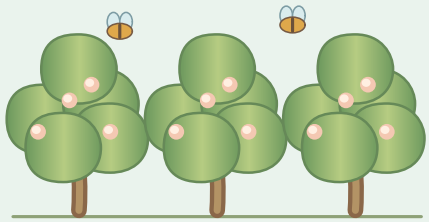
REVIEW HABIT

Explain a pathway aloud using the diagram before checking the definitions.

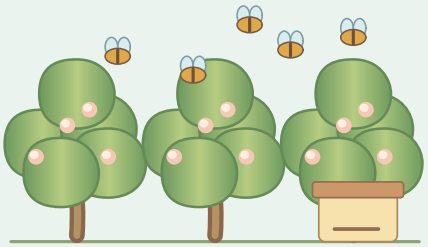
PRACTICE

NCERT Folder

APPLES AND POLLINATORS / TWO ORCHARD SITES



Orchard A
Naturally occurring pollinators



Orchard B
Managed bee colonies present

The reported comparison suggests a hypothesis. A matched study would control for site differences.

Think it over

A grower wants one batch of banana plants with uniform traits and another crop with new combinations of traits. Explain when asexual propagation would help and when a controlled sexual cross would help.

Pause and ponder

A pupil claims that a fruit has already formed as soon as pollen lands on the stigma. Follow the stages from pollen landing to seed and fruit development. Explain what still has to happen.

Revise, reflect, refine

A class tests five sugar concentrations using pollen from a single flower species. Some grains form visible tubes. Identify the independent variable, a measurable dependent variable and three variables to keep constant. Explain why repeating the investigation improves confidence in the result.

Apply what you learned: Apples and bees

An orchard that keeps managed bee colonies shows higher fruit set and less premature fruit drop than an orchard relying on natural pollinators. The orchards are in different places. Suggest a reason for the observed difference and one alternative explanation. Design a comparison that separates the effect of bee colonies from the effect of location.

EVIDENCE CHECK

A comparison can support a hypothesis. When orchard locations differ, soil, weather or tree variety could also affect the result.

PRACTICE

Exam-Style Questions

Choose the best answer

1. Which structure produces pollen? A Stigma B Anther C Ovary D Sepal.
2. Which cells normally have 23 chromosomes in humans? A Uterine lining cells B A zygote C Gametes D Most skin cells.
3. Which method keeps a branch attached to the parent while it grows roots? A Grafting B Tissue culture C Budding D Layering.

Short explanations

4. Explain why a pollen grain on the stigma has not yet fertilised the egg.
5. Explain why spore formation allows Rhizopus to spread efficiently.
6. Distinguish ovulation, fertilisation and implantation by giving the site and outcome of each event.

Long answer

7. Compare self-pollination with cross-pollination. Give one example of a flower feature that supports each method and explain how repeated self-pollination may affect variation.
8. Trace the human embryo's route from fertilisation to implantation. Explain the roles of the placenta and umbilical cord later in pregnancy.

Case question

9. A grower has a mango tree with excellent fruit and a second tree with a strong root system. The grower wants to combine these benefits. Name the technique, identify the scion and rootstock, and explain why the join has to be aligned and protected.

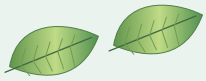
DIAGRAM PROMPTS / USE WITH QUESTIONS 7–9



Q7 / Flowers
How might pollen move?



Q8 / Early development
Where would the embryo travel?



Q9 / Mango plants
How could these traits be combined?

Describe what the diagrams show and connect your answer to the question.

EXAM TECHNIQUE
In an explain question, link the structure, the process and the result.

PRACTICE

Self Assessment

Recall and compare

1. Define asexual reproduction and name two examples that use different methods.
2. Explain why a farmer might choose tissue culture for a large batch of planting stock.
3. Name the flower structure that becomes the seed and the structure that usually becomes the fruit.

Assertion and reason

4. Assertion: A human zygote normally has 46 chromosomes.
Reason: Human sperm and eggs normally contribute 23 chromosomes each.
Decide whether each statement is true and whether the reason explains the assertion.

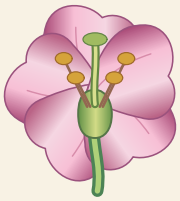
Interpret a case

5. A moist slice of bread kept in a warm cupboard develops fuzzy growth. Under a microscope, upright threads end in round sacs. Identify the threads and sacs. Explain how the sacs help the mould spread and predict how refrigeration would affect its growth.

Check understanding

6. A student says ovulation always occurs on day 14. Explain the basis of the day-14 model and why the statement is incorrect.
7. Name one contraception method that also lowers the risk of many STIs. Explain why hormonal pills do not offer that additional protection.

SELF-CHECK / THREE CONCEPTS TO EXPLAIN



Q3 / Flower structures
What becomes seed and fruit?



Q5 / Bread mould
Name the stalk and sac



Q6 / Cycle timing
Is day 14 a rule or a model?

Use the illustrations to support your reasoning.

DIAGRAM PRACTICE

Sketch a flower in longitudinal section. Label the anther, filament, stigma, style, ovary and an ovule.

PRACTICE

Answers and Source Notes

NCERT Folder

Uniform bananas can be multiplied by vegetative propagation. Controlled crosses can create new trait combinations. Pollen must germinate, grow a tube and deliver a gamete before fertilisation. Sugar concentration is changed; percentage of grains forming tubes is measured. Use the same pollen source, temperature and observation time. In the apple case, compare matched plots at the same sites, with and without managed bees.

Exam-style questions

1–3. B, C, D. **4.** Pollination is pollen transfer. Fertilisation requires a tube and gamete fusion. **5.** Many small spores disperse and germinate in favourable conditions. **6.** Ovary releases egg; fusion in oviduct forms zygote; embryo embeds in uterine lining. **7.** Same plant versus different individuals; enclosed stamens can favour selfing and feathery stigmas can favour cross-transfer by wind. Repeated selfing can reduce diversity. **8.** Divisions occur en route to uterus; implantation occurs in endometrium. Placenta exchanges materials, and the cord connects it to the foetus. **9.** Grafting; fruit tree supplies scion and strong-rooted tree supplies rootstock. Contact between growing tissues and protection of the join support healing.

Self assessment

1. One parent, no gamete fusion; examples include yeast budding and Rhizopus spores. **2.** Controlled culture multiplies uniform plantlets from a small sample. **3.** Ovule → seed; ovary → fruit. **4.** Both statements are true and the reason explains the assertion. **5.** Hyphae and sporangia; spores disperse, and refrigeration slows mould growth. **6.** Day 14 is an illustration for a 28-day cycle; cycles and ovulation timing vary. **7.** Condoms reduce the risk of many STIs; pills do not form a barrier.

Sources

Theory sequence adapted from the supplied *Arihant All in One Science Class 9*, "Reproduction: How Life Continues", pp. 344–367. Explanations, diagrams and practice questions have been prepared for Rama's Academy.
NCERT exemplar: angiosperm ovule · OpenStax: fungal structures · Endotext: implantation
WHO: contraception · WHO: abortion care · ACOG: adolescent cycles · National Health Mission: family planning · PCPNDT Act 1994, sections 3A, 5 and 6.

ANSWER KEY / WHY THE ZYGOTE HAS 46



Human sperm and egg normally contribute one chromosome set each.